

Z = stdev for pop have

T = stdev for pop no, just sample

Confidence interval

Two way, back to back.

- Z test and T test uses the same formula, **critical value** is positive only, that is what the formula is made for, so calculate right tail critical value

$$\bar{x} - \frac{z_{\alpha/2} \cdot \sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + \frac{z_{\alpha/2} \cdot \sigma}{\sqrt{n}}$$

σ is a standard deviation, for T, use sample, for Z, use pop

if $n \geq 30$, use Z test right away because sample is large enough

Chi square

$$\frac{(n-1)s^2}{X_{\alpha/2, n-1}^2} \leq \sigma^2 \leq \frac{(n-1)s^2}{X_{1-\alpha/2, n-1}^2}$$

The formula is about denominator so it is switched, for a larger/smaller value

Confidence bound

One way, very extreme

Remember, Z, T, and X are critical value

Either, for upper confidence bound, for z and t

$$\mu \leq u = \bar{x} + \frac{z_{\alpha} \cdot \sigma}{\sqrt{n}}$$

Or, for lower confidence bound, for z and t

$$\bar{x} - \frac{z_{\alpha} \cdot \sigma}{\sqrt{n}} = l \leq \mu$$

or **chi square**, upper conf bound, remember, X is critical value

s^2 is sample variance

WARNING : alpha means ALPHA FROM THE RIGHT

$$\frac{(n-1)s^2}{X_{\alpha, n-1}^2} \leq \sigma^2$$

And lower con bound

$$\sigma^2 \leq \frac{(n-1)s^2}{X_{1-\alpha, n-1}^2}$$

Chi square test

Is not symmetric so two tail test for P value, needs to determine first whether which side of the graph is the test stats stay, then find the extreme value according to that side and multiply by 2. **center of the graph is degree of freedom**

Proportion, sample prop and pop prop

CI

If $\hat{p}$ is the proportion of observations in a random sample of size n that belongs to a class of interest, an approximate $100(1 - \alpha)\%$ confidence interval on the proportion p of the population that belongs to this class is

$$\hat{p} - z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} \leq p \leq \hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

where $z_{\alpha/2}$ is the upper $\alpha/2$ percentage point of the standard normal distribution.

CB

The approximate $100(1 - \alpha)\%$ lower and upper confidence bounds are

$$\hat{p} - z_{\alpha} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} \leq p \quad \text{and} \quad p \leq \hat{p} + z_{\alpha} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

respectively.

Reverse engineer, how large n we require to get that we can make sure that the error is below 0.05 with 95% confidence

$$n = \left(\frac{z_{\alpha/2}}{E} \right)^2 p(1 - p)$$

P is **sample** proportion, Z is also left tail as always. The error, well, depends

For upper bound on n , change $p(1-p)$ to be 0.25

Test statistics for Z test

$\bar{X}$ = sample mean

μ_0 = hypothesis

σ = **pop stdev**

n = sample count

$$z_0 = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

Test statistics for T test

$\bar{X}$ = sample mean

μ_0 = hypothesis

s = sample stdev

n = sample count

$$t_0 = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$$

Test statistics for Chi Square χ^2

n = sample count

s^2 is sample variance

σ^2 is hypothesized variance

$$\chi_0^2 = \frac{(n - 1)s^2}{\sigma_0^2}$$

Test statistics for Z test (proportion)

$\hat{P}$ = sample proportion

p_0 = hypothesized proportion

n = sample count

$$Z_0 = \frac{\hat{P} - p_0}{\sqrt{p_0(1 - p_0)/n}}$$

- **population** mean : μ
- sample mean : $\bar{X}$
- **population** variance : σ^2
- sample variance : s^2

in THESE INTERVALS AND BOUND, it has to be right tail critical value, to get the positive value

Two sample Z test : known pop stdev

Null hypothesis: $H_0: \mu_1 - \mu_2 = \Delta_0$

Test statistic:
$$Z_0 = \frac{\bar{X}_1 - \bar{X}_2 - \Delta_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Alternative Hypotheses	P-Value	Rejection Criterion For for Fixed-Level Tests
$H_1: \mu_1 - \mu_2 \neq \Delta_0$	Probability above $ z_0 $ and probability below $- z_0 $, $P = 2[1 - \Phi(z_0)]$	$z_0 > z_{\alpha/2}$ or $z_0 < -z_{\alpha/2}$
$H_1: \mu_1 - \mu_2 > \Delta_0$	Probability above z_0 , $P = 1 - \Phi(z_0)$	$z_0 > z_\alpha$
$H_1: \mu_1 - \mu_2 < \Delta_0$	Probability below z_0 , $P = \Phi(z_0)$	$z_0 < -z_\alpha$

If $\bar{x}_1$ and $\bar{x}_2$ are the means of independent random samples of sizes n_1 and n_2 from two independent normal populations with known variances σ_1^2 and σ_2^2 , respectively, a **100(1 - α)% confidence interval for $\mu_1 - \mu_2$** is

$$\bar{x}_1 - \bar{x}_2 - z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \leq \mu_1 - \mu_2 \leq \bar{x}_1 - \bar{x}_2 + z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

where $z_{\alpha/2}$ is the upper $\alpha/2$ percentage point of the standard normal distribution.

- Choice of Sample Size

$$n = \left(\frac{z_{\alpha/2}}{E} \right)^2 (\sigma_1^2 + \sigma_2^2)$$

- One-Sided Confidence Bounds

- Upper Confidence Bound

$$\mu_1 - \mu_2 \leq \bar{x}_1 - \bar{x}_2 + z_\alpha \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

- Lower Confidence Bound

$$\bar{x}_1 - \bar{x}_2 - z_\alpha \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \leq \mu_1 - \mu_2$$

Two sample pooled t-test : pop stdev are unknown, equal pop stdev

The **pooled estimator** of σ^2 , denoted by S_p^2 , is defined by

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

Null hypothesis: $H_0: \mu_1 - \mu_2 = \Delta_0$

Test statistic: $T_0 = \frac{\bar{X}_1 - \bar{X}_2 - \Delta_0}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$

Alternative Hypothesis	P-Value	Rejection Criterion for Fixed-Level Tests
$H_1: \mu_1 - \mu_2 \neq \Delta_0$	Probability above $ t_0 $ and probability below $- t_0 $	$t_0 > t_{\alpha/2, n_1 + n_2 - 2}$ or $t_0 < -t_{\alpha/2, n_1 + n_2 - 2}$
$H_1: \mu_1 - \mu_2 > \Delta_0$	Probability above t_0	$t_0 > t_{\alpha, n_1 + n_2 - 2}$
$H_1: \mu_1 - \mu_2 < \Delta_0$	Probability below t_0	$t_0 < -t_{\alpha, n_1 + n_2 - 2}$

If $\bar{x}_1, \bar{x}_2, s_1^2$, and s_2^2 are the sample means and variances of two random samples of sizes n_1 and n_2 , respectively, from two independent normal populations with unknown but equal variances, then a $100(1 - \alpha)\%$ confidence interval on the difference in means $\mu_1 - \mu_2$ is

$$\begin{aligned} \bar{x}_1 - \bar{x}_2 - t_{\alpha/2, n_1 + n_2 - 2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \\ \leq \mu_1 - \mu_2 \leq \bar{x}_1 - \bar{x}_2 + t_{\alpha/2, n_1 + n_2 - 2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \end{aligned}$$

where $s_p = \sqrt{[(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2]/(n_1 + n_2 - 2)}$ is the pooled estimate of the common population standard deviation, and $t_{\alpha/2, n_1 + n_2 - 2}$ is the upper $\alpha/2$ percentage point of the t distribution with $n_1 + n_2 - 2$ degrees of freedom.

Two sample Welch's t-test : pop stdev are unknown, unequal pop stdev

If $H_0: \mu_1 - \mu_2 = \Delta_0$ is true, the statistic

$$T_0^* = \frac{\bar{X}_1 - \bar{X}_2 - \Delta_0}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

$$v = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$$

V is degree of freedom

If $\bar{x}_1$, $\bar{x}_2$, s_1^2 , and s_2^2 are the means and variances of two random samples of sizes n_1 and n_2 , respectively, from two independent normal populations with unknown and unequal variances, an approximate $100(1 - \alpha)\%$ confidence interval on the difference in means $\mu_1 - \mu_2$ is

$$\bar{x}_1 - \bar{x}_2 - t_{\alpha/2, v} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \leq \mu_1 - \mu_2 \leq \bar{x}_1 - \bar{x}_2 + t_{\alpha/2, v} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

where v is given by Equation 10-16 and $t_{\alpha/2, v}$ is the upper $\alpha/2$ percentage point of the t distribution with v degrees of freedom.

Paired t test : two matched or paired measurement, same item, different testing under two different conditions or methods.

- same subject twice, before and after **same number of samples**

Null hypothesis: $H_0: \mu_D = \Delta_0$

Test statistic: $T_0 = \frac{\bar{D} - \Delta_0}{S_D/\sqrt{n}}$

Alternative Hypothesis	P-Value	Rejection Region for Fixed-Level Tests
$H_1: \mu_D \neq \Delta_0$	Sum of the probability above t_0 and the probability below $-t_0$,	$t_0 > t_{\alpha/2, n-1}$ or $t_0 < -t_{\alpha/2, n-1}$
$H_1: \mu_D > \Delta_0$	Probability above t_0	$t_0 > t_{\alpha, n-1}$
$H_1: \mu_D < \Delta_0$	Probability below t_0	$t_0 < -t_{\alpha, n-1}$

D bar is mean of differences of two data, Sd is likewise

If $\bar{d}$ and s_D are the sample mean and standard deviation of the difference of n random pairs of normally distributed measurements, a $100(1 - \alpha)\%$ confidence interval on the difference in means $\mu_D = \mu_1 - \mu_2$ is

$$\bar{d} - t_{\alpha/2, n-1} s_D/\sqrt{n} \leq \mu_D \leq \bar{d} + t_{\alpha/2, n-1} s_D/\sqrt{n}$$

where $t_{\alpha/2, n-1}$ is the upper $\alpha/2\%$ point of the t distribution with $n - 1$ degrees of freedom.

Two variance F-test : measure variance of two data, equal or not

Null hypothesis: $H_0: \sigma_1^2 = \sigma_2^2$

Test statistic: $F_0 = \frac{S_1^2}{S_2^2}$

Alternative Hypotheses	Rejection Criterion
$H_1: \sigma_1^2 \neq \sigma_2^2$	$f_0 > f_{\alpha/2, n_1-1, n_2-1}$ or $f_0 < f_{1-\alpha/2, n_1-1, n_2-1}$
$H_1: \sigma_1^2 > \sigma_2^2$	$f_0 > f_{\alpha, n_1-1, n_2-1}$
$H_1: \sigma_1^2 < \sigma_2^2$	$f_0 < f_{1-\alpha, n_1-1, n_2-1}$

If s_1^2 and s_2^2 are the sample variances of random samples of sizes n_1 and n_2 , respectively, from two independent normal populations with unknown variances σ_1^2 and σ_2^2 , then a $100(1 - \alpha)\%$ confidence interval on the ratio σ_1^2/σ_2^2 is

$$\frac{s_1^2}{s_2^2} f_{1-\alpha/2, n_2-1, n_1-1} \leq \frac{\sigma_1^2}{\sigma_2^2} \leq \frac{s_1^2}{s_2^2} f_{\alpha/2, n_2-1, n_1-1}$$

where $f_{\alpha/2, n_2-1, n_1-1}$ and $f_{1-\alpha/2, n_2-1, n_1-1}$ are the upper and lower $\alpha/2$ percentage points of the F distribution with $n_2 - 1$ numerator and $n_1 - 1$ denominator degrees of freedom, respectively. A confidence interval on the ratio of the standard deviations can be obtained by taking square roots in Equation 10-33.

For two tail test, the center of the distribution is 1

Two proportion Z-test : measure proportion of data

Null hypothesis: $H_0: p_1 = p_2$

Test statistic: $Z_0 = \frac{\hat{P}_1 - \hat{P}_2}{\sqrt{\hat{P}(1 - \hat{P})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$

Alternative Hypothesis	P-Value	Rejection Criterion for Fixed-Level Tests
$H_1: p_1 \neq p_2$	Probability above $ z_0 $ and probability below $- z_0 $. $P = 2[1 - \Phi(z_0)]$	$z_0 > z_{\alpha/2}$ or $z_0 < -z_{\alpha/2}$
$H_1: p_1 > p_2$	Probability above z_0 . $P = 1 - \Phi(z_0)$	$z_0 > z_\alpha$
$H_1: p_1 < p_2$	Probability below z_0 . $P = \Phi(z_0)$	$z_0 < -z_\alpha$

$\hat{P}$ is total P

$$r = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sqrt{\left(\sum x^2 - \frac{(\sum x)^2}{n}\right) \left(\sum y^2 - \frac{(\sum y)^2}{n}\right)}}$$

NOT IMPORTANT

CORREL(data1, data2) = r

Simple linear regression

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i\right)^2}{n}$$

$$S_{xy} = \sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) = \sum_{i=1}^n x_i y_i - \frac{\left(\sum_{i=1}^n x_i\right)\left(\sum_{i=1}^n y_i\right)}{n}$$

Therefore, the least squares estimates of the slope and intercept are

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} =$$

and

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} =$$

The fitted simple linear regression model (with the coefficients reported to three decimal places) is

A $100(1 - \alpha) \%$ **prediction interval on a future observation** Y_0 at the value x_0 is given by

$$\begin{aligned} \hat{y}_0 - t_{\alpha/2, n-2} \sqrt{\hat{\sigma}^2 \left[1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right]} \\ \leq Y_0 \leq \hat{y}_0 + t_{\alpha/2, n-2} \sqrt{\hat{\sigma}^2 \left[1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right]} \end{aligned}$$

The value $\hat{y}_0$ is computed from the regression model $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$.

Y_0 and X_0 are the exact point

$$SS_E = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$\hat{y}_i$ is the predicted Y. Real Y - predicted Y

$$\hat{\sigma}^2 = \frac{SS_E}{n - 2}$$

Multivariable regression

$X_1 * X_2, X_1^2, X_2^2$ Extra column

Input in data analysis, Y is y, x is all x, including extra column

Look at P value, select those that is under 0.05 (95% CI), significant enough

P value < alpha, reject H_0

Two regression always intersects at the same point

$$r = \sqrt{b_{yx} \times b_{xy}}$$

Correlation of two regression, of x based y, and of y based x

When $r = 0$ (no correlation) the best predictor of one variable, by another, is its mean, because that is all we have

Design experiment

Conjecture– the original hypothesis that motivates the experiment.

Treatment combination	Design Factors			Thickness				Total
	A	B	AB					
(1)	-	-	+	14.037	14.165	13.972	13.907	56.081
a	+	-	-	14.821	14.757	14.843	14.878	59.299
b	-	+	-	13.880	13.860	14.032	13.914	55.686
ab	+	+	+	14.888	14.921	14.415	14.932	59.156

a, b, ab, (1) are total from here, this is for effect estimator for A B AB

$$A = \bar{y}_{A+} - \bar{y}_{A-} = \frac{a + ab}{2n} - \frac{b + (1)}{2n} = \frac{1}{2n} [a + ab - b - (1)]$$

$$B = \bar{y}_{B+} - \bar{y}_{B-} = \frac{b + ab}{2n} - \frac{a + (1)}{2n} = \frac{1}{2n} [b + ab - a - (1)]$$

$$AB = \frac{ab + (1)}{2n} - \frac{a + b}{2n} = \frac{1}{2n} [ab + (1) - a - b]$$

$$SS_A = \frac{[a + ab - b - (1)]^2}{4n}$$

$$SS_B = \frac{[b + ab - a - (1)]^2}{4n}$$

$$SS_{AB} = \frac{[ab + (1) - a - b]^2}{4n}$$

k = number of variables (A, B). n is number of tests (4) from green table above

$$\text{Coefficient} = \frac{\text{effect}}{2}$$

$$se(\text{Effect}) = \sqrt{\frac{2\hat{\sigma}^2}{N/2}} = \sqrt{\frac{\hat{\sigma}^2}{n2^{k-2}}}$$

$$se(\text{Coefficient}) = \frac{1}{2} \sqrt{\frac{2\hat{\sigma}^2}{N/2}} = \frac{1}{2} \sqrt{\frac{\hat{\sigma}^2}{n2^{k-2}}}$$

$$t \text{ ratio} = \frac{\text{effect}}{se(\text{effect})} = \frac{\text{coefficient}}{se(\text{coefficient})}$$

$$\hat{\sigma}^2 = \text{mean square error}$$

$$2^k(n - 1) = \text{residual degrees of freedom}$$

σ^2 is average of variances of each variable test == (4 test for 2 variables, (1), a b ab) (average(variance of each row))

(standard error (effect))² = variance (effect)

residual degrees of freedom = degree of freedom

Effect Estimator	Estimated Standard Error	t Ratio	P-Value	*Sig
0.8360	0.0721	11.5960	0.0000	*
-0.0672	0.0721	-0.9328	0.3693	
0.0315	0.0721	0.4369	0.6699	

effect estimator comes from effect estimator that was found earlier, t ratio has formula, P value is a two tail of T, an area, only those under alpha is accepted, significant enough

Average =	14.389	AVERAGE(E3:H6)		
Coef_A =	0.418	=B46/2		
Coef_B =	-0.034			
Coef_AB =	0.016			
if factors are not significant enough to be taken in to calculation				
Prediction Model : $y = 0.418 A + 14.389$				

Average here is average of every test, represent Y intercept

=y is thickness, that is why mean is like that, and we are trying to get a predictor equation

Qualitative : discrete

Quantitative : continuous

1. Find effect estimator (there is a formula)
2. Find standard effect error (use variance, k number of trials, and n number of variables)
3. find t ratio ef/se
4. find P value (use T ratio and find two tail of it, degree of freedom can be found using formula)
5. identify significance value
6. find coefficient
7. prediction model with significance only, with Y intercept as mean of all

2^k design experiment

contrast -> effect estimator -> standard error -> t-ratio -> P value -> significant model

$$\text{Effect} = \frac{\text{Contrast}}{n2^{k-1}}$$

$$SS = \frac{(\text{Contrast})^2}{n2^k}$$

for effect, the k-1 formula is affected for fractional design, since the no. of real variable is reduced

	A	B	C	AB	AC	BC	ABC
	-16	-16	-16	16	16	16	-16
	22	-22	-22	-22	-22	22	22
	-20	20	-20	-20	20	-20	20
	27	27	-27	27	-27	-27	-27
	-21	-21	21	21	-21	-21	21
	23	-23	23	-23	23	-23	-23
	-18	18	18	-18	-18	18	-18
	30	30	30	30	30	30	30
Contrast	27	13	7	11	1	-5	9
Effect	3.375	1.625	0.875	1.375	0.125	-0.625	1.125
Sum Square	45.563	10.563	3.0625	7.5625	0.0625	1.5625	5.0625

contrast is to combine the entire column above it.

Focus on the sum square, it is the determiner.

Coefficient is effect/2.

Statistical process control

S.D. of sample mean = stdev/sqrt(n)

3-sigma control limit

- UCL = mean + 3* s.d. of sample mean
- CL = mean
- LCL = mean - 3* s.d. of sample mean

- Actual value can exceed this limit

Table X Factors for Constructing Variables Control Charts

n^*	Factor for Control Limits						
	$\bar{X}$ Chart			R Chart		S Chart	
	A_1	A_2	d_2	D_3	D_4	c_4	n
2	3.760	1.880	1.128	0	3.267	0.7979	2
3	2.394	1.023	1.693	0	2.575	0.8862	3
4	1.880	.729	2.059	0	2.282	0.9213	4
5	1.596	.577	2.326	0	2.115	0.9400	5
6	1.410	.483	2.534	0	2.004	0.9515	6
7	1.277	.419	2.704	.076	1.924	0.9594	7
8	1.175	.373	2.847	.136	1.864	0.9650	8
9	1.094	.337	2.970	.184	1.816	0.9693	9
10	1.028	.308	3.078	.223	1.777	0.9727	10
11	.973	.285	3.173	.256	1.744	0.9754	11
12	.925	.266	3.258	.284	1.716	0.9776	12
13	.884	.249	3.336	.308	1.692	0.9794	13
14	.848	.235	3.407	.329	1.671	0.9810	14
15	.816	.223	3.472	.348	1.652	0.9823	15
16	.788	.212	3.532	.364	1.636	0.9835	16
17	.762	.203	3.588	.379	1.621	0.9845	17
18	.738	.194	3.640	.392	1.608	0.9854	18
19	.717	.187	3.689	.404	1.596	0.9862	19
20	.697	.180	3.735	.414	1.586	0.9869	20
21	.679	.173	3.778	.425	1.575	0.9876	21
22	.662	.167	3.819	.434	1.566	0.9882	22
23	.647	.162	3.858	.443	1.557	0.9887	23
24	.632	.157	3.895	.452	1.548	0.9892	24
25	.619	.153	3.931	.459	1.541	0.9896	25

* $n > 25$: $A_1 = 3/\sqrt{n}$ where n = number of observations in sample.

Sample	Viscosity	UCL	CL	LCL
1	378	505.2120	480.1000	454.9880
2	438	505.2120	480.1000	454.9880
3	487	505.2120	480.1000	454.9880
4	515	505.2120	480.1000	454.9880

For graph creation purpose

X-bar Control Chart			
UCL =	36.6666	$UCL = \bar{\bar{x}} + A_2\bar{r} \quad CL = \bar{\bar{x}} \quad LCL = \bar{\bar{x}} - A_2\bar{r}$	
CL =	33.3200		
LCL =	29.9734		
R Control Chart			
UCL =	12.2612	$UCL = D_4\bar{r} \quad CL = \bar{r} \quad LCL = D_3\bar{r}$	
CL =	5.8000		
LCL =	0.0000		
S Control Chart			
UCL =	4.8985	$UCL = \bar{s} + 3\frac{\bar{s}}{c_4}\sqrt{1 - c_4^2} \quad CL = \bar{s} \quad LCL = \bar{s} - 3\frac{\bar{s}}{c_4}\sqrt{1 - c_4^2}$	
CL =	2.3451		
LCL =	-0.2084		
X-bar Control Chart from S			
UCL =	36.6671	$UCL = \bar{\bar{x}} + 3\frac{\bar{s}}{c_4\sqrt{n}} \quad CL = \bar{\bar{x}} \quad LCL = \bar{\bar{x}} - 3\frac{\bar{s}}{c_4\sqrt{n}}$	
CL =	33.3200		
LCL =	29.9729		

The **moving range** (MR) is defined as the absolute difference between two successive observations:

$$MR_i = |x_i - x_{i-1}|$$

The center line and upper and lower control limits for a control chart for individuals are

$$UCL = \bar{\bar{x}} + 3\frac{\overline{mr}}{d_2} = \bar{\bar{x}} + 3\frac{\overline{mr}}{1.128}$$

$$CL = \bar{\bar{x}}$$

$$LCL = \bar{\bar{x}} - 3\frac{\overline{mr}}{d_2} = \bar{\bar{x}} - 3\frac{\overline{mr}}{1.128}$$

and for a control chart for moving ranges

$$UCL = D_4\overline{mr} = 3.267\overline{mr}$$

$$CL = \overline{mr}$$

$$LCL = D_3\overline{mr} = 0$$

The **process capability ratio** (PCR) is

$$PCR = \frac{USL - LSL}{6\sigma}$$

$$PCR_k = \min \left[\frac{USL - \mu}{3\sigma}, \frac{\mu - LSL}{3\sigma} \right]$$

customer's promise, specification

$$z = \frac{\text{value} - \text{mean}}{\sigma}$$

can it fit?

does it skew near border?

$P(X < LSL)$	
$= P(Z < (90 - 107)/1.5)$	
$= P(Z < -11.33) =$	0.0000
$P(X > USL)$	
$= P(Z > (110 - 107)/1.5)$	
$= P(Z > 2) =$	0.0228

Probability that it goes out the border, turning normal var into z and use norm.s.dist

The center line and upper and lower control limits for the P chart are

$$UCL = \bar{p} + 3 \sqrt{\frac{\bar{p}(1 - \bar{p})}{n}} \quad CL = \bar{p} \quad LCL = \bar{p} - 3 \sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}$$

where $\bar{p}$ is the observed value of the average fraction defective.

proportion-related

The center line and upper and lower control limits on the U chart are

$$UCL = \bar{u} + 3 \sqrt{\frac{\bar{u}}{n}} \quad CL = \bar{u} \quad LCL = \bar{u} - 3 \sqrt{\frac{\bar{u}}{n}}$$

where $\bar{u}$ is the average number of defects per unit.

n is no. of sample per batch, 5 boards are inspected every hour

	A	B	C	D	E	F
1	Target for process mean = 99					
2	Sample	Concentration (x_i)	$x_i - 99$	$s_i = (x_i - 99) + s_{i-1}$	Moving Average (4)	Moving Average (6)
3	1	102.0	3.0	3.0		
4	2	94.8	-4.2	-1.2		
5	3	98.3	-0.7	-1.9		
6	4	98.4	-0.6	-2.5	98.4	
7	5	102.0	3.0	0.5	98.4	
8	6	98.5	-0.5	0.0	99.3 	99.0